Algebra 1 Name___Answer Key________
Unit 2 Date____________________
Lesson 3 Homework Period___________________

1. Tamer was simplifying the expression and made an error. His steps are shown. Find and fix the mistake.

$$\left(\frac{-2xy \cdot 3x^2}{5x^4y}\right)^2$$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{-6x^6y^2}{5x^8y^2}$

Step 3: $\frac{-6}{5x^2}$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{36x^6y^2}{25x^8y^2}$

Step 3: $\frac{36}{25x^2}$

2. Write a note to Tamer explaining his error.

Apply Power of a Product Law of Exponents also to numerical coefficients.

For questions 3-5, find an equivalent expression using the laws of exponents.

3. $\frac{-6a^7b^9}{(-2ab^3)^5}$

$$\frac{-6a^7b^9}{-32a^5b^{15}} = \frac{3a^2}{16b^6}$$

4. $\frac{(9x^8y)\cdot(-2x^2y^5)}{3x^5y^{10}}$

$$\frac{-18x^{10}y^6}{3x^5y^{10}} = \frac{-6x^5}{y^4}$$

5. $(\frac{-3mn^3}{4m^4 \cdot 2m^2n^5})^3$

$$\frac{-27m^3n^9}{512m^{18}n^{15}} = \frac{-27}{512m^{15}n^6}$$

6. Prove $2^{3x+5} = 32(8)^x$ by using the laws of exponents.

$2^{3x+5} = 32(8)^x$
$(2^{3x})(2^5) = 32(8)^x$
$32(2^3)^x = 32(8)^x$
$32(8)^x = 32(8)^x$

7. Prove $27^{6a-1} = \frac{(3^{6a})^3}{9 \cdot 3}$ by using the laws of exponents.

$27^{6a-1} = \frac{(3^{6a})^3}{9 \cdot 3}$
$(27^{6a})(27^{-1}) = \frac{(3^{6a})^3}{9 \cdot 3}$
$[(3^3)^{6a}](27^{-1}) = \frac{(3^{6a})^3}{9 \cdot 3}$
$\frac{3^{18a}}{27} = \frac{(3^{6a})^3}{9 \cdot 3}$
$\frac{(3^{6a})^3}{9 \cdot 3} = \frac{(3^{6a})^3}{9 \cdot 3}$

8. Write an equivalent expression for $(-5.2)^{2x^3} \cdot (-5.2)^{x^2} \cdot (-5.2)^{-5}$.

$(-5.2)^{2x^3+x^2-5}$

9. Use the law of exponents to write an equivalent expression for 4^{3-2c}.

$(4^3)(4^{-2c}) = \frac{4^3}{4^{2c}}$

10. Use the law of exponents to write an equivalent expression for 3.5^{3x+7}.

$(3.5^{3x})(3.5^7) = (3.5^x)^3(3.5^7) = (3.5^3)^x(3.5^7)$